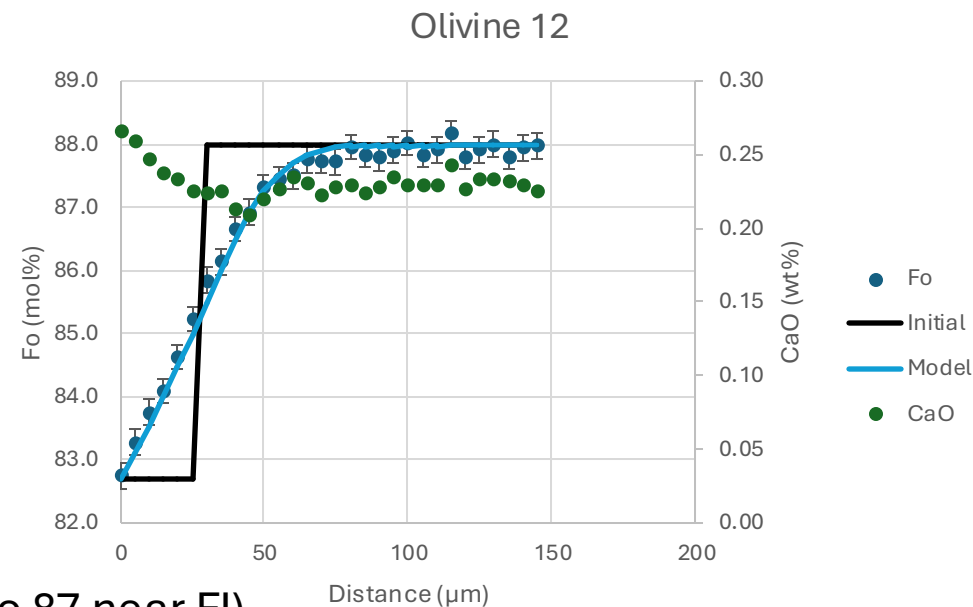
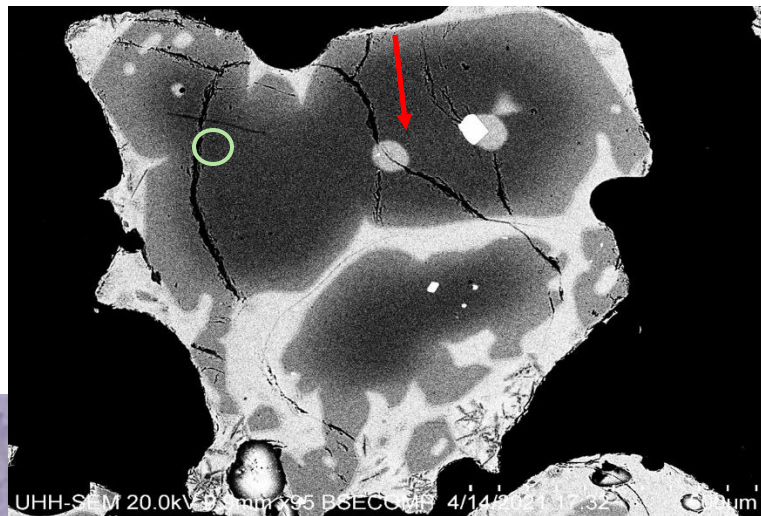
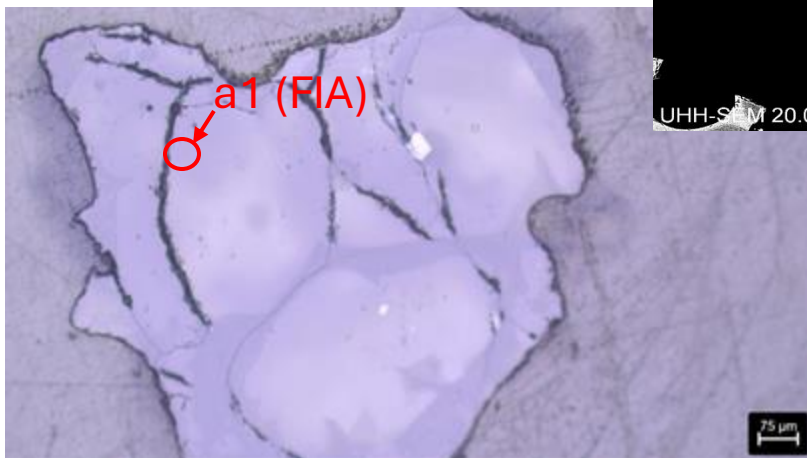


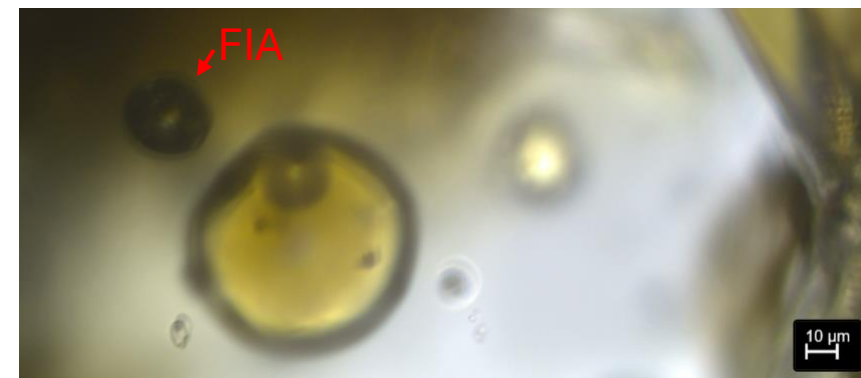
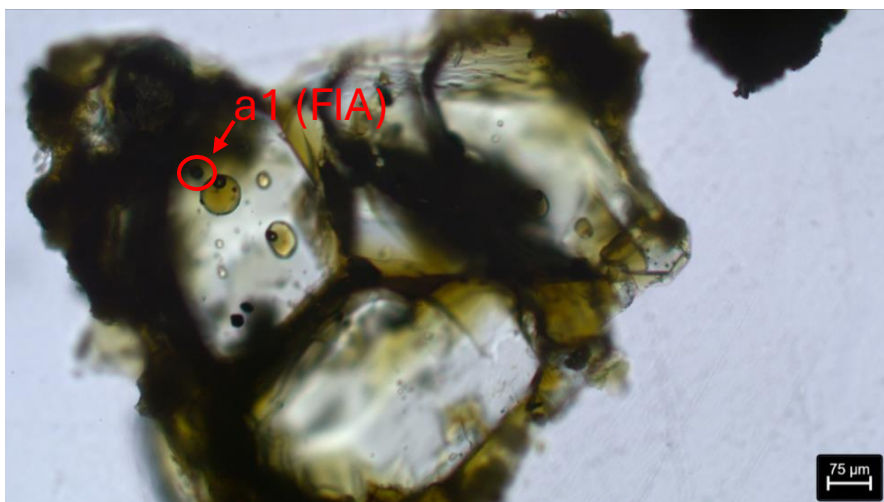
K20_c001

EDS spots

K20_c001_a1: associated to FIA
Diad +so2, nothing else on broad



Not the exact same grain (Fo 87 near FI),
but same cluster and the cluster seems to predate the Fe-Mg
disequilibrium rim. Grain boundaries are connected and they are higher Fo
than the low Fo rim. Skeletal growth? Undeformed

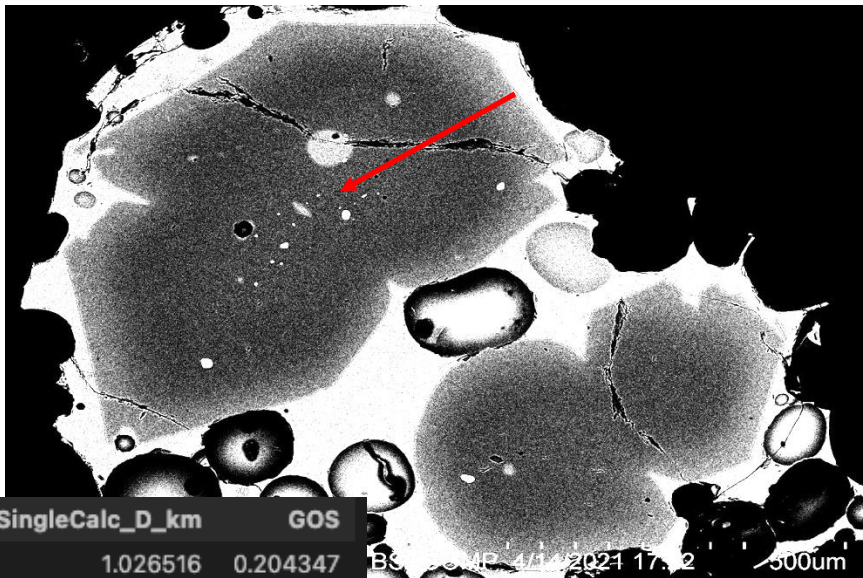


filename	Density g/cm3	Fo_Ol	SingleCalc_D_km	GOS
310 K20_c001_a1_FIA	0.115712	0.872405	1.709659	0.136585

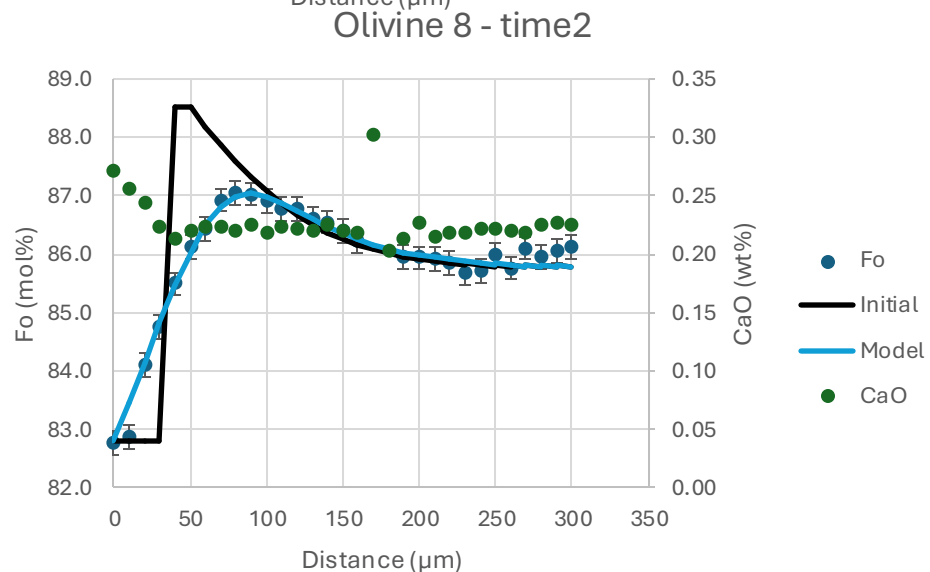
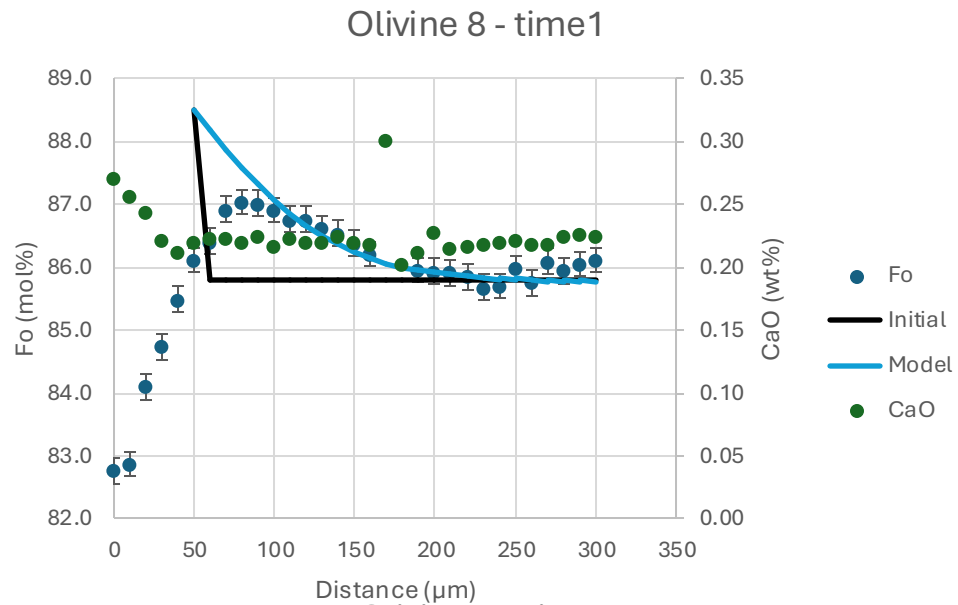
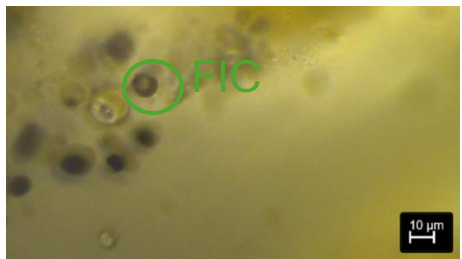
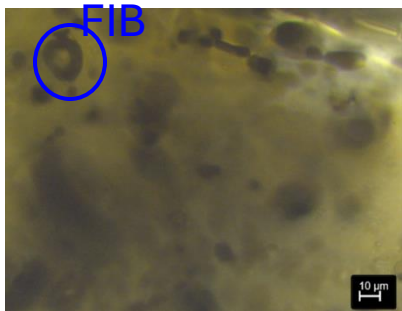
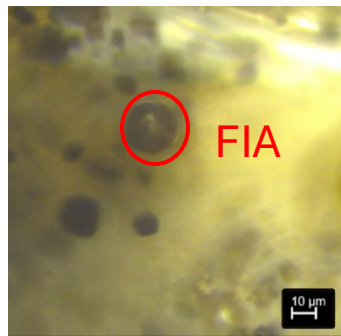
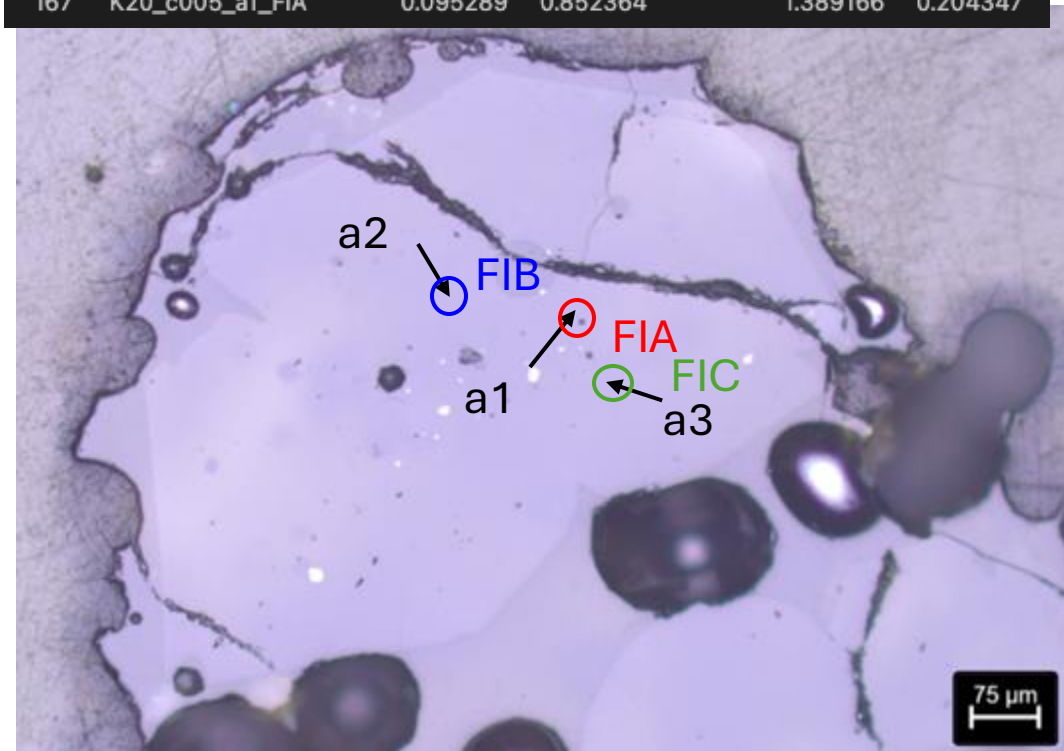
K20_c005
Diads+SO2
EDS spots

K20_c005_a1 (FIA)
K20_c005_a2 (FIB)
K20_c005_a3 (FIC)

Complex zoned “shoulder”
from Lynn 2024, undeformed



	filename	Density g/cm3	Fo_OI	SingleCalc_D_km	GOS
77	K20_c005_a2_FIB	0.068034	0.860823	1.026516	0.204347
123	K20_c005_a3_FIC	0.084521	0.862001	1.255884	0.204347
167	K20_c005_a1_FIA	0.095289	0.852364	1.389166	0.204347



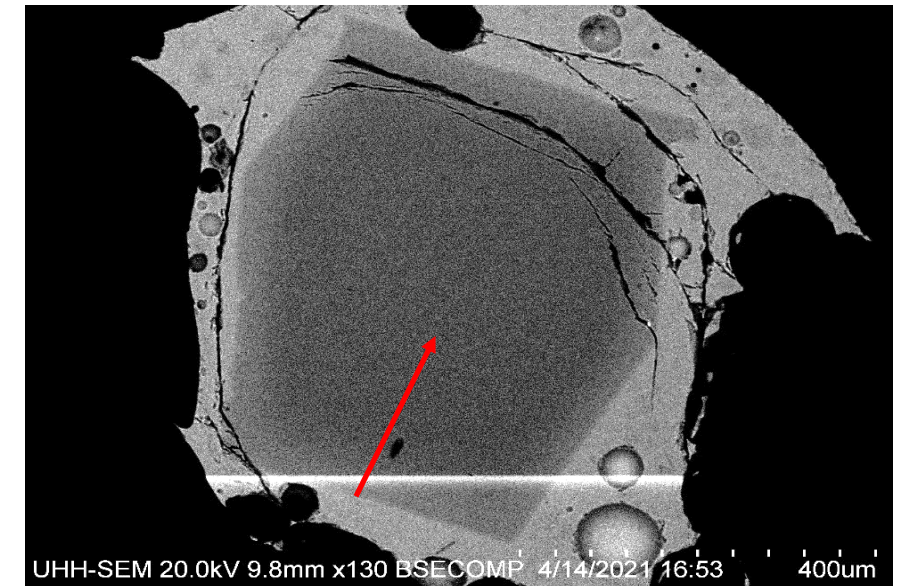
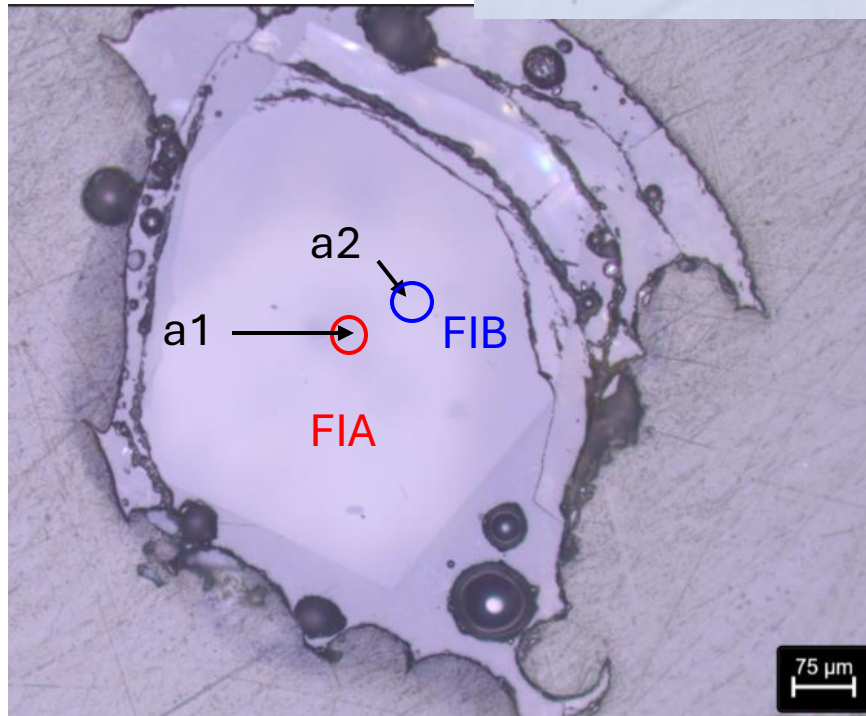
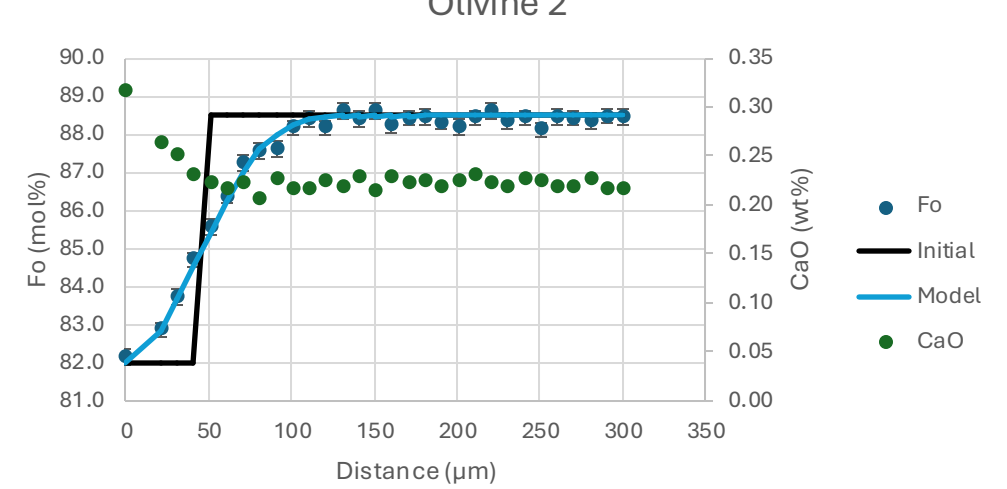
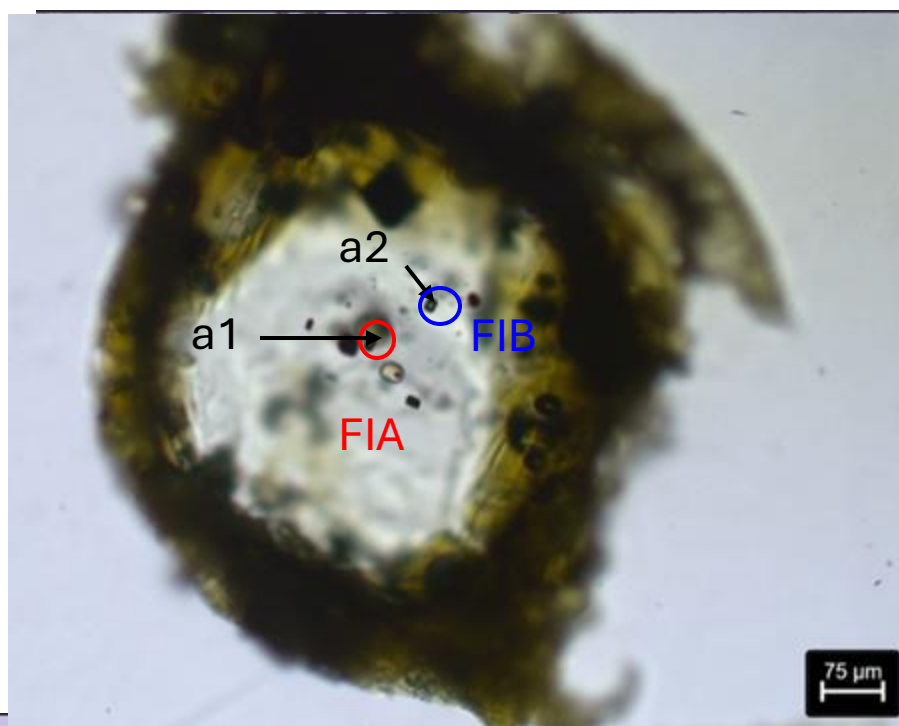
K20_c008

Diads+SO2

EDS spots

K20_c008_a1 (FIA)

K20_c008_a2 (FIB)



Normal zoning, high Fo core, the FI are in the high Fo core, undeformed

	filename	Density g/cm3	Fo_OI	SingleCalc_D_km	GOS
69	K20_c008_a1_FIA	0.064505	0.880733	1.002641	0.128481
101	K20_c008_a2_FIB	0.077172	0.880834	1.182734	0.128481

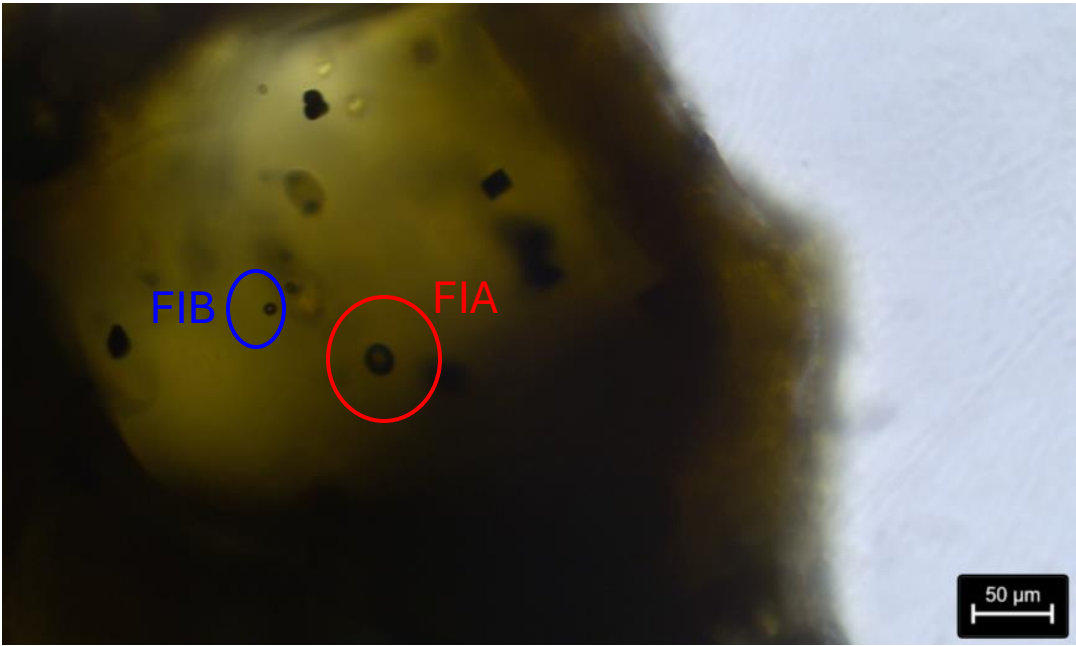
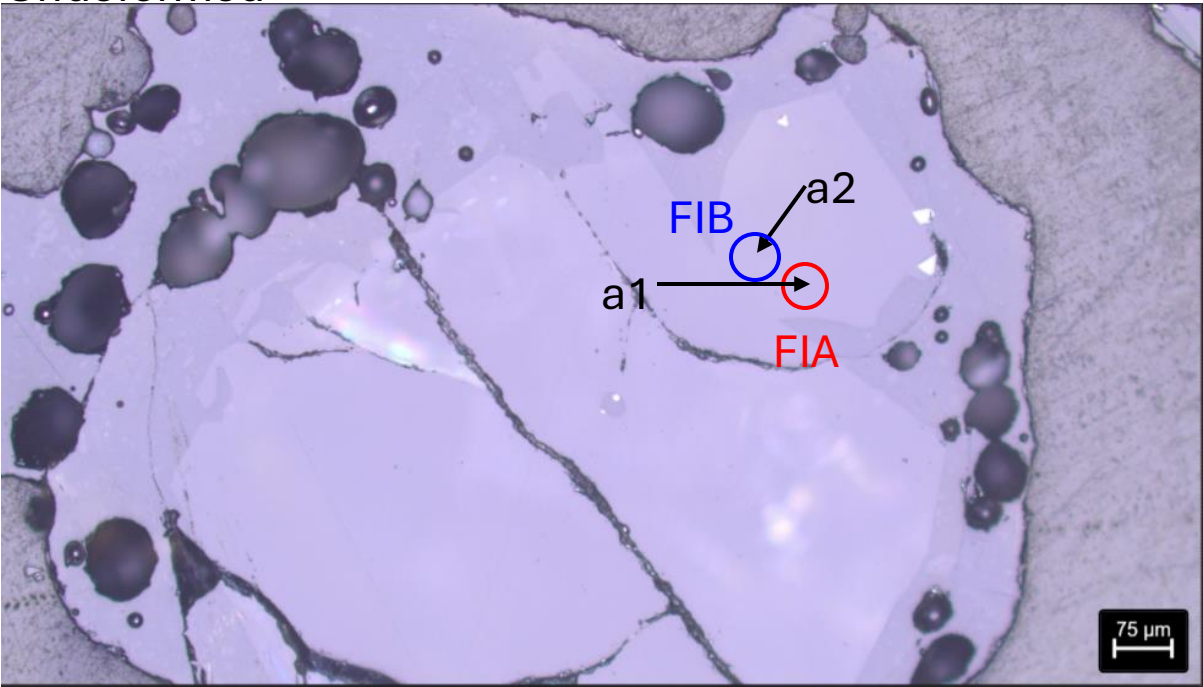
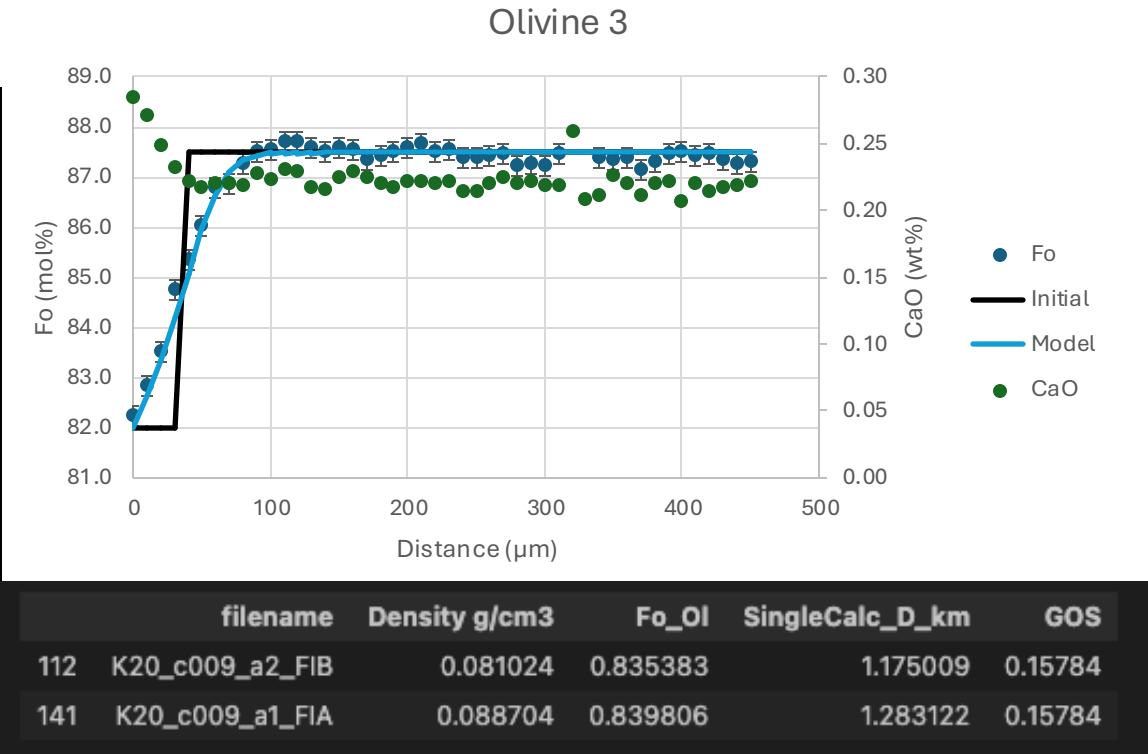
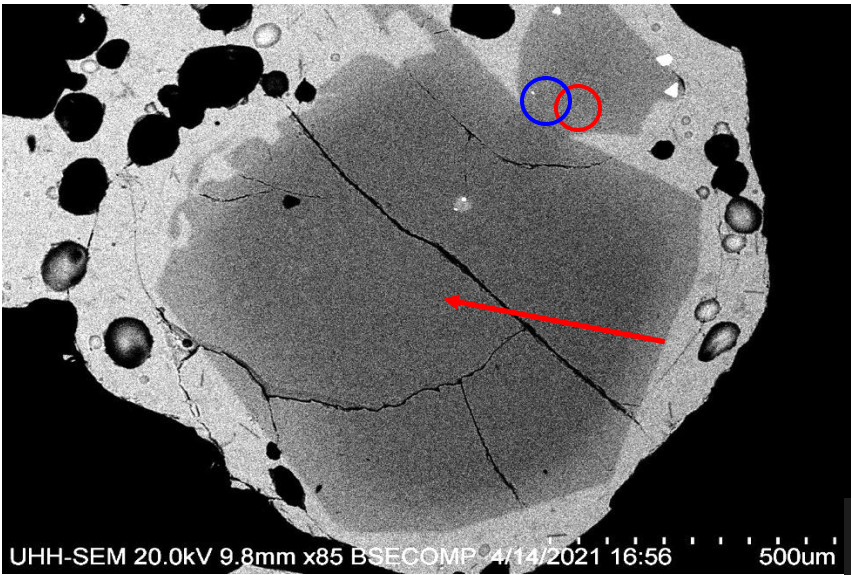
K20_c009

EDS spots

K20_c009_a1 (FIA)

K20_c009_a2 (FIB)

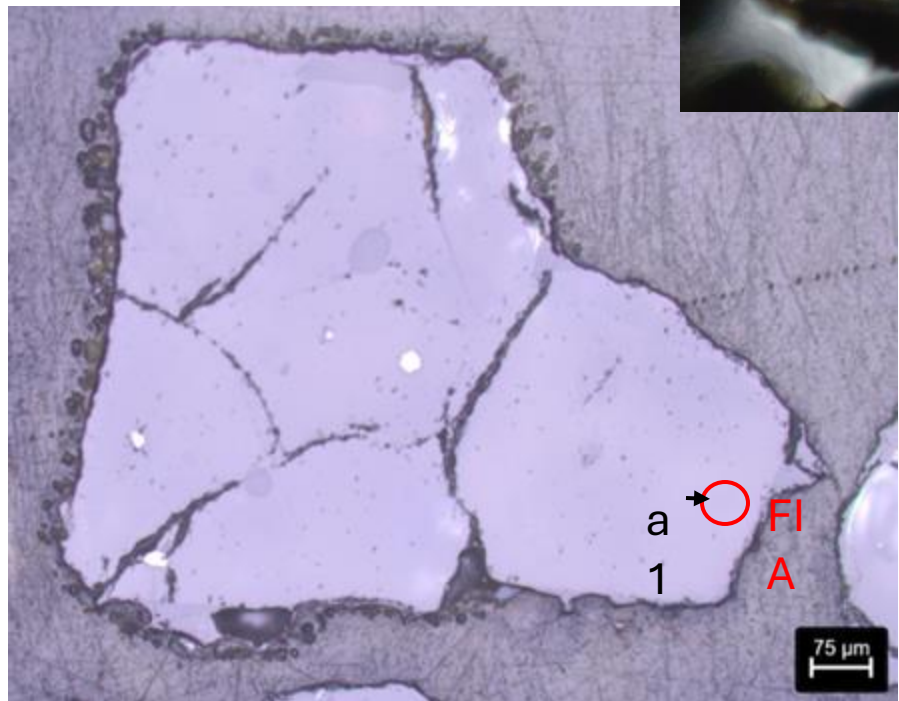
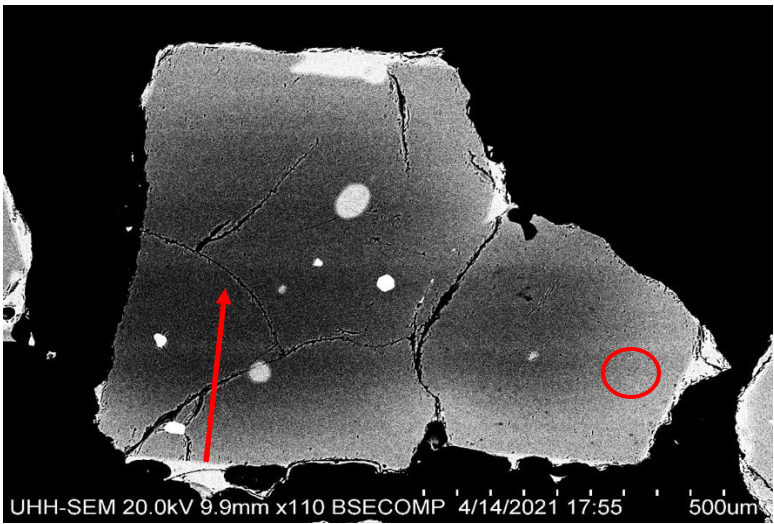
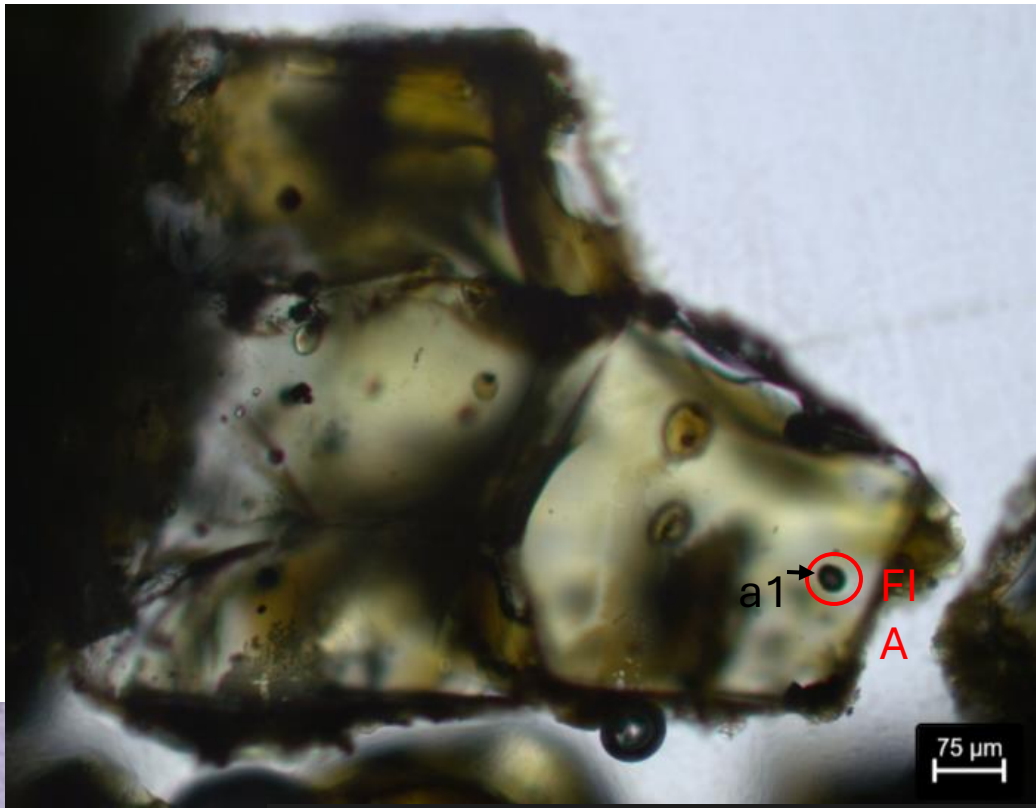
Kendra's is the big crystal, normally zoned. The smaller crystal is where the FI are, at the boundary. Doesn't look zoned and is low Fo
Undeformed



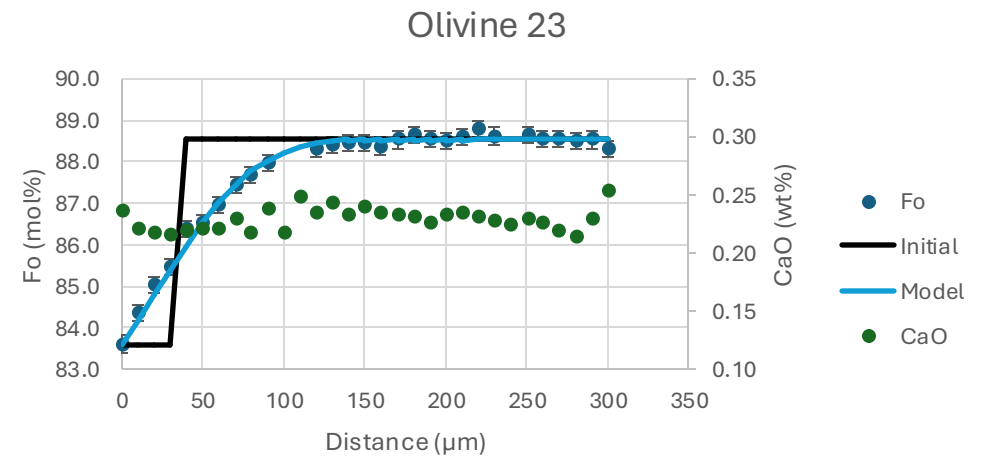
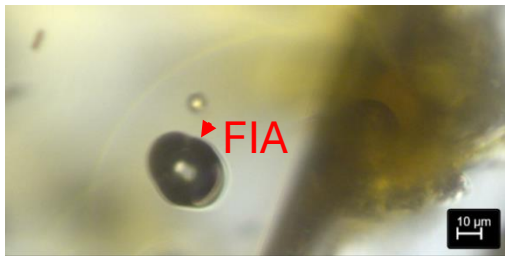
K20_c015
Has a melt film
Diad

EDS spots
K20_c015_a1 (FIA)

Not the exact same subgrain
but same cluster. Again looks
like cluster predates the Fe-Mg
diseq event. It's in the low Fo
rim and is low P.
Undeformed



	filename	Density g/cm3	Fo_Ol	SingleCalc_D_km	GOS
142	K20_c015_a1_FIA	0.08911	0.830208	1.277673	0.244097



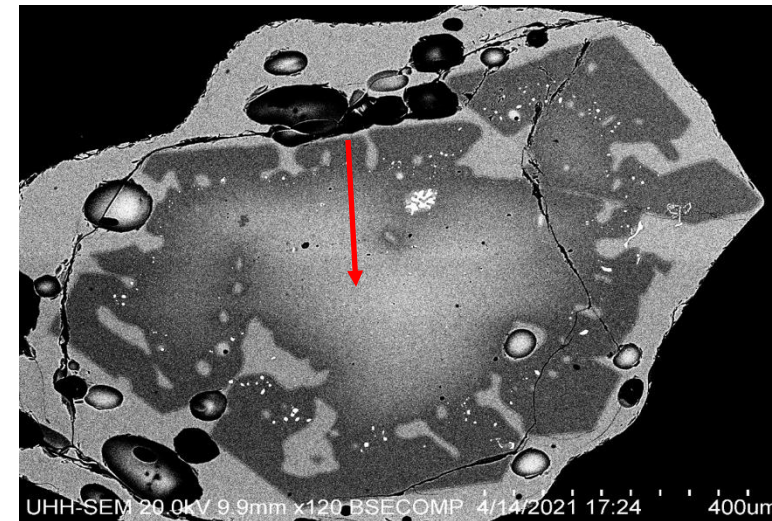
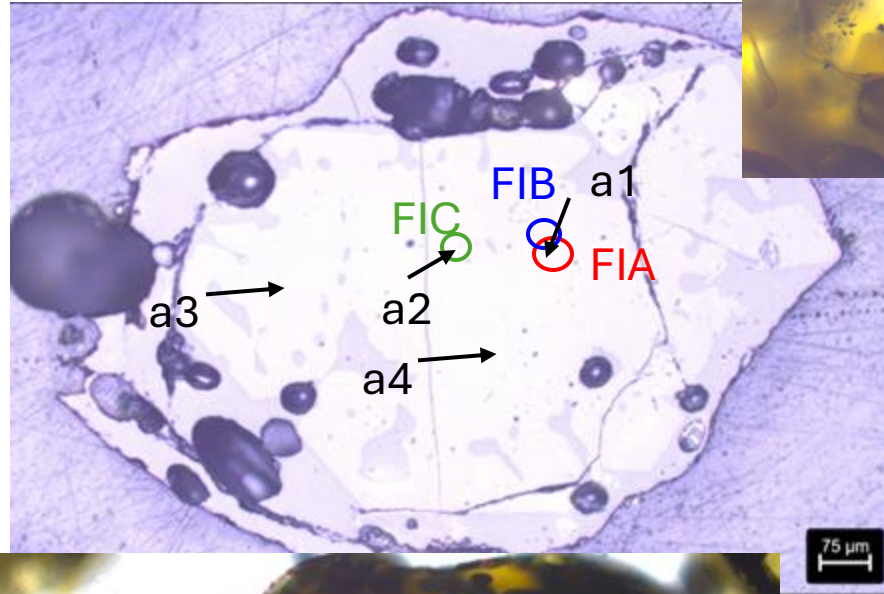
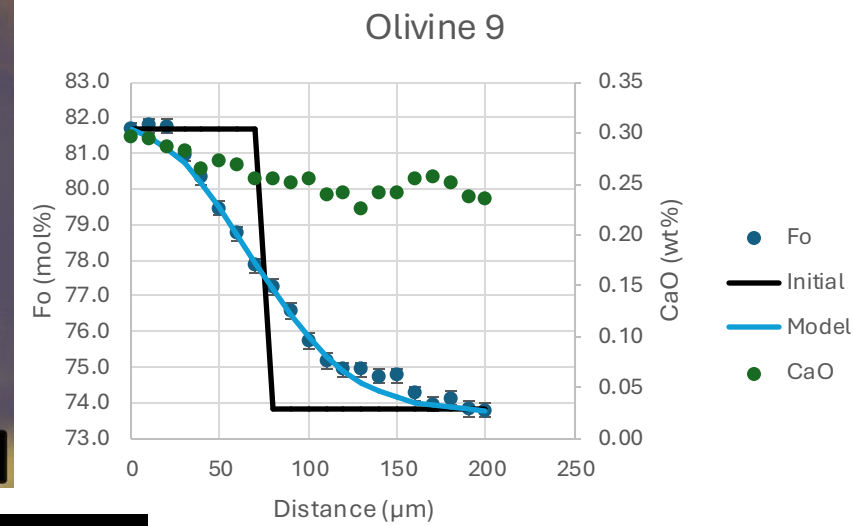
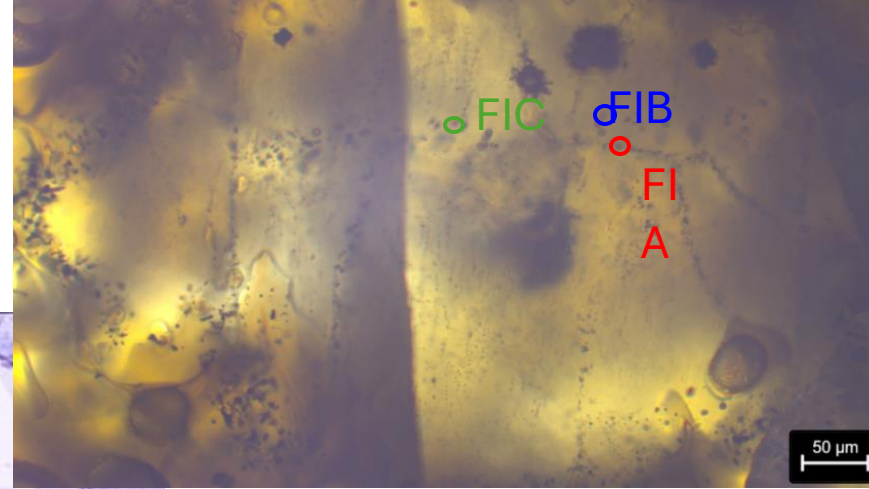
K20_c017, No diads

EDS spots

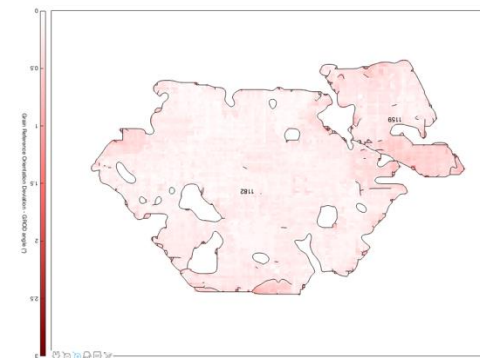
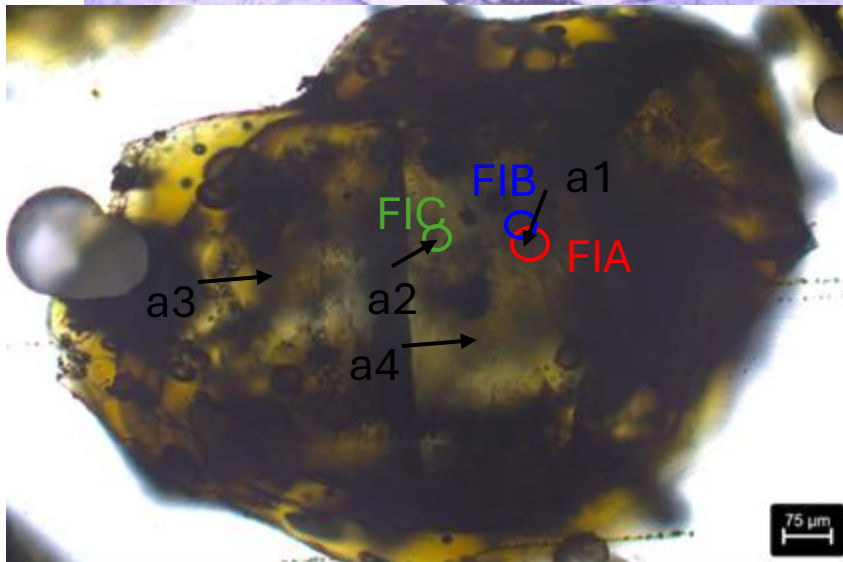
K20_c017_a1 (FIA, FIB)

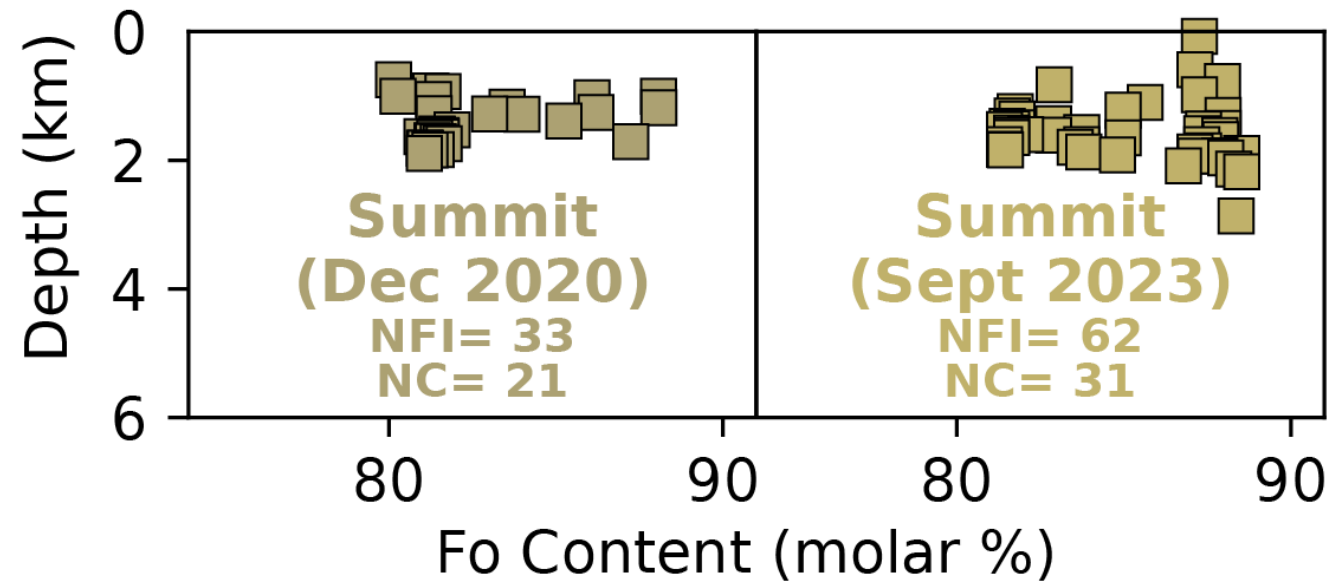
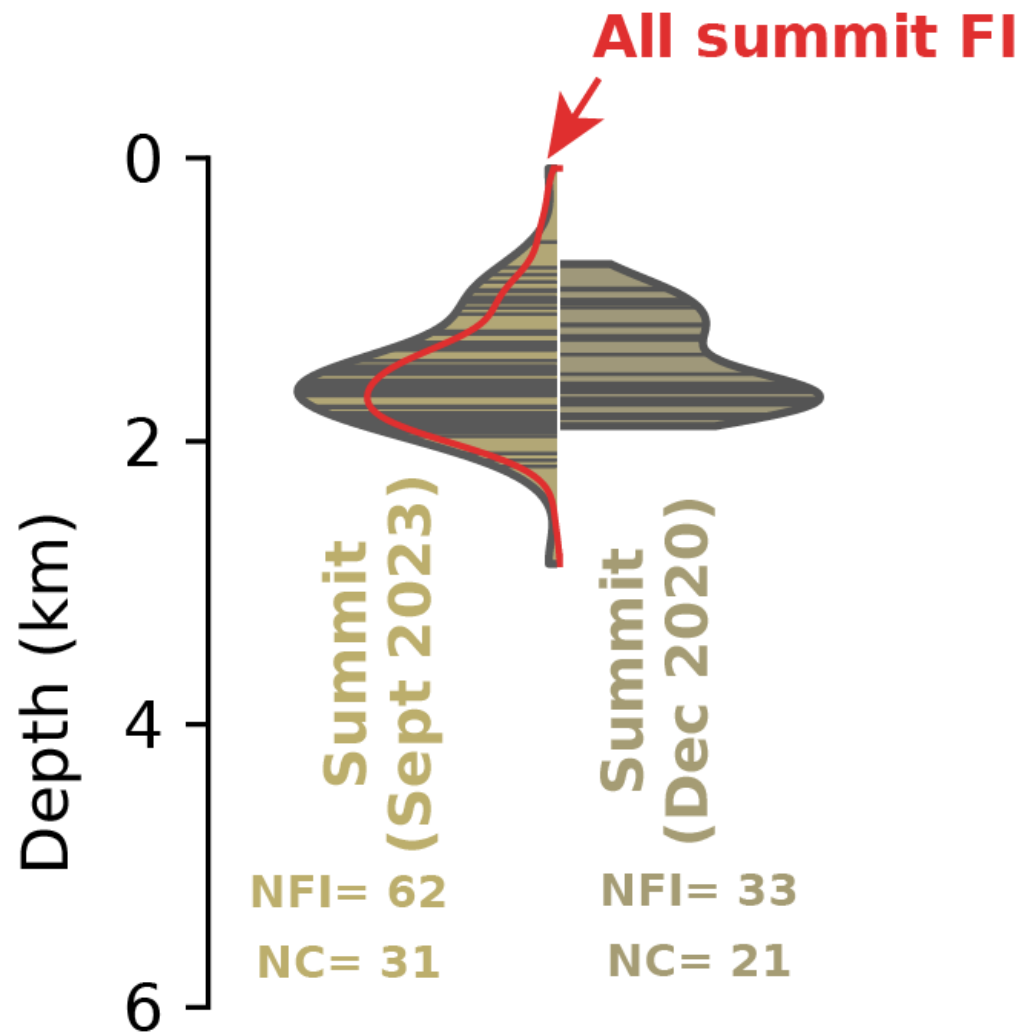
K20_c017_a2 (FIC)

K20_c017_a3 and a4 (other spots)



Reverse zoned crystal from Kendra. FI are in the low Fo core but they are all secondary. What happened here? Are the secondaries related to the dissolution event? they had no diads though....Or are they older remnants? Either way, no diads. Undeformed





What if the recharge source wasn't SC but rather the 1.7 ~ km region, and it recharged into ~1km. The summit FI are lightly bi modal and K20 is bimodal at 1.17 , 1.69 km. K23 is also bimodal, but the upper at 0.9 km.

If the high MgO magmas are at 1.7 km, grow olivine and then move to 1.2 before erupting, that wouldn't require the direct involvement of deeper magmas and explain that the high Fo olivines like 001 have low P FI.

Although it doesn't really work with 008...which is ~1.1 km and is Fo 88...